VIRTUAL CLOTHES TRY-ON SYSTEM

A Project Synopsis Submitted in Partial Fulfillment of the

Bachelor of Technology Degree

In

Information Technology

Submitted by

Arohi Dwivedi (2201220130034)

Under the guidance of

Er. Vijay Shukla



Shri Ramswaroop Memorial College of Engineering & Management, Lucknow
Affiliated to Dr. APJ Abdul Kalam Technical University, Uttar Pradesh, Lucknow

TABLE OF CONTENT

1. Introduction
1.1. Background……. 1
1.2. Significance of Virtual Clothes Try On……..……. 1

2. Literature Survey
2.1. Introduction……..…… 2
2.2. Existing Commercial Solutions
2.2.1. Amazon and Zeekit Virtual Try-On…….. 2
2.2.2. Myntra & Flipkart Virtual Try-On…….……... 2
2.2.3. Snapchat & AR Filters……..…….2
2.3. Academic and Research-Based Approaches ……..……. 2

3. Problem Definition
3.1. Problem Statement……..…… 4
3.2. Objectives……..… 4
3.3. Proposed Methodology……..……..…… 5

4. Virtual Clothes Try On System
4.1. Introduction……..…… 6
4.2. System Architecture……...…….. 6
4.3. System Features……..……. 7

5. Requirement Analysis
5.1. Hardware Requirements……..……...…….. 8
5.2. Software Requirements…… 8
5.3. Feasibility Study
5.3.1. Technical Feasibility……...……. 9
5.3.2. Economic Feasibility……...…… 9
5.3.3. Operational Feasibility……..…….. 9

6. Module Description
6.1. Modules……..…….10
6.2. Module Integration……..13

7. Data Flow Diagram……...…14

8. Applications, Advantages, Limitations & Future Scope……...…….....….... 15

9. References……...…… 18

CHAPTER 1: INTRODUCTION

1.1 Background

With the rise of digital commerce, fashion retail has undergone a fundamental transformation. Consumers increasingly prefer online platforms for convenience, variety, and accessibility. However, a key drawback persists: the inability to physically try on clothing before purchase. Traditional product images or size charts often fail to provide an accurate sense of fit and appearance. This gap results in **uncertainty for consumers and high return rates for retailers**, which in turn impacts customer trust and operational costs.

To address this challenge, researchers and developers have explored **virtual try-on technologies**. Virtual try-on systems aim to allow users to visualize clothing on themselves without physically wearing it. Early attempts relied on 2D overlay methods, which simply placed clothing images on top of user photos. These approaches often lacked realism and failed to adapt clothing to individual body shapes.

Recent advancements in **computer vision and deep learning** have significantly improved the realism of virtual try-on systems. Models such as **CP-VTON (Clothing Image-based Virtual Try-On Network)** use **pose estimation, image segmentation, and garment warping** to generate highly realistic results. These systems analyze the user's posture and body structure, align the selected clothing accordingly, and produce an output image that closely mimics reality.

1.2 Significance of Virtual Clothes Try-On

The concept of virtual clothes try-on is particularly significant for several reasons:

- **E-commerce Enhancement**: Helps consumers make informed purchasing decisions by providing realistic previews.
- **Customer Satisfaction**: Reduces uncertainty about size, fit, and appearance, leading to fewer product returns.
- **Sustainability**: Decreases logistical waste by minimizing unnecessary returns and shipping.
- **Fashion Design & Retail**: Assists designers and brands in showcasing collections digitally without requiring physical samples.

Accessibility: Provides users with diverse body types the ability to check clothing compatibility more accurately.

CHAPTER 2: LITERATURE SURVEY

2.1 Introduction

A literature survey provides an overview of existing technologies, commercial solutions, and academic research relevant to a given project. For this work, the focus is on **virtual try-on systems in fashion technology**. These systems aim to allow users to visualize how garments would look on them without physical trials. This chapter highlights commercial attempts, open-source frameworks, and research advancements that shape the foundation of the **Virtual Clothes Try-On** project.

2.2 Existing Commercial Solutions

2.2.1 Amazon and Zeekit Virtual Try-On

Amazon acquired Zeekit, a virtual try-on platform that overlays clothing on user-uploaded photos. While it provides a visual approximation, it relies heavily on 2D image overlay and lacks realistic garment draping and texture simulation.

2.2.2 Myntra & Flipkart Virtual Try-On

These Indian e-commerce platforms experimented with basic try-on features for accessories and simple apparel. However, the results often appear unrealistic due to poor alignment with body poses and limited adaptability to diverse body types.

2.2.3 Snapchat & AR Filters

Social media platforms like Snapchat and Instagram have developed augmented reality (AR) filters for clothing and accessories. These filters provide interactive try-on experiences but are optimized for entertainment rather than precise garment fitting.

Limitations of Commercial Systems:

- Heavy reliance on 2D overlays without advanced garment warping.
- Limited adaptability to complex garments or varying poses.
- More focused on marketing than practical fit visualization.

2.3 Academic and Research-Based Approaches

2.3.1 CP-VTON (Clothing Image-based Virtual Try-On Network)

- One of the most influential research models in virtual try-on.
- Uses pose-guided person generation and a geometric matching module (GMM) to align clothes with the user's body.
- Generates realistic try-on images but requires high computational resources.

2.3.2 GAN-Based Approaches

- Generative Adversarial Networks (GANs) have been widely adopted for virtual try-on.
- GAN-based models generate high-quality, photo-realistic results by training a generator-discriminator pair.
- Examples include VITON and MG-VTON, which improve garment alignment and realism.

2.3.3 Segmentation and Warping Methods

- Some approaches rely on image segmentation to separate body parts and garments before alignment.
- Warping techniques adjust clothing images to match the user's body shape.
- These methods struggle with preserving garment details, especially in textured or patterned clothing.

2.3.4 3D Virtual Try-On Systems

- 3D-based methods model the human body in three dimensions for greater realism.
- While accurate, they require expensive hardware, depth sensors, or full-body scans, making them less practical for everyday e-commerce applications.

CHAPTER 3: PROBLEM DEFINITION

3.1 Problem Statement

The growth of online shopping has created convenience and accessibility for millions of consumers. However, one major challenge persists in the **fashion and apparel industry**: customers cannot physically try on clothing before purchase. This leads to uncertainty about **fit, size, and appearance**, often resulting in dissatisfaction, increased return rates, and loss of trust in digital retail platforms.

Existing solutions attempt to address this issue through **2D overlays, augmented reality filters, or 3D body modelling**, but these approaches suffer from limitations such as poor realism, lack of adaptability, and high computational costs. Advanced research models like **CP-VTON (Clothing Image-based Virtual Try-On Network)** show promise but require further adaptation for practical deployment in e-commerce.

Thus, the **core problem** is:

"How can we design an affordable, scalable, and realistic virtual try-on system that enables customers to visualize clothing on themselves using only a photograph and a clothing image?"

3.2 Objectives

The project aims to achieve the following objectives:

1. **Clothing Visualization**
    - Generate realistic images of users wearing selected garments using deep learning models.
2. **Pose Estimation and Body Alignment**
    - Accurately identify body posture and key points for proper garment alignment.
3. **Clothing Warping and Fitting**
    - Warp garment images to match user body shape and pose without losing texture details.
4. **Segmentation and Blending**
    - Perform human body segmentation to isolate regions (torso, arms, etc.) and blend garments seamlessly.
5. **System Accessibility**
    - Ensure that the solution runs on consumer-grade hardware using open-source frameworks.
6. **E-commerce Integration**
    - Develop a prototype that can be extended into online retail platforms to reduce return rates and improve user confidence.

3.3 Proposed Methodology

The proposed **Virtual Clothes Try-On System** is designed as a multi-step process integrating computer vision and deep learning techniques:

1. **Input Acquisition**
   - User uploads an image of themselves.
   - Clothing image is uploaded or selected from a catalogue.
2. **Preprocessing**
   - Normalize input images (resize, colour adjustments).
   - Apply background removal if necessary.
3. **Pose Estimation**
   - Detect body landmarks using pose estimation models (e.g., OpenPose, MediaPipe).
   - Extract skeletal structure and orientation of the user.
4. **Human Parsing / Segmentation**
   - Perform segmentation of user image into distinct regions (torso, arms, background).
   - Ensure clothing overlays only the appropriate body regions.
5. **Garment Warping (CP-VTON Module)**
   - Warp clothing image using a **Geometric Matching Module (GMM)** to align it with the user's body.
   - Preserve texture, patterns, and garment shape.
6. **Image Synthesis**
   - Blend the warped garment with the segmented user image using deep generative models.
   - Ensure smooth transitions between clothing and body parts.
7. **Output Generation**
   - Produce a realistic image of the user wearing the chosen garment.
   - Display results for preview and validation.

CHAPTER 4: VIRTUAL CLOTHES TRY-ON SYSTEM

4.1 Introduction

The **Virtual Clothes Try-On System (VCTOS)** is the central framework of this project. Its primary purpose is to enable users to digitally try on garments using only a personal image and a clothing image. Unlike simple overlay methods, the proposed system leverages **deep learning and computer vision** to ensure that clothing aligns realistically with the user's **pose, body structure, and proportions**.

The architecture of VCTOS is designed in a **layered and modular manner** to ensure flexibility, scalability, and ease of integration with e-commerce platforms. Each layer focuses on a specific functionality, from **input processing** to **garment warping** and **output visualization**.

4.2 System Architecture

The architecture of VCTOS can be divided into the following layers:

1. **Input Layer**
    - Accepts two inputs: (a) user image and (b) garment image.
    - Ensures compatibility in image format and resolution.
2. **Preprocessing Layer**
    - Normalizes and resizes input images.
    - Removes background noise to isolate the user's body.
3. **Pose Estimation Layer**
    - Uses body key point detection (OpenPose/MediaPipe) to extract skeletal posture.
    - Aligns garment position according to body pose.
4. **Segmentation Layer**
    - Divides the user's body into distinct regions (torso, arms, legs).
    - Ensures clothing overlays correctly without overlapping unwanted areas.
5. **Garment Warping Layer**
    - Utilizes **Geometric Matching Module (GMM)** in CP-VTON.
    - Warps garment to fit the user's body dimensions and pose.
6. **Image Synthesis Layer**
    - Uses generative models to blend warped clothing with the user image.
    - Preserves garment textures, patterns, and fabric details.
7. **Output Layer**
    - Produces the final try-on image.
    - Displays results in a user interface for evaluation.

4.3 System Features

- **Realistic Visualization –** Produces photo-like results instead of crude overlays.
- **Texture Preservation –** Maintains fabric patterns and garment details after warping.
- **Adaptability –** Works with a variety of garments (tops, dresses, jackets).
- **Scalability –** Modular design allows integration with mobile apps and retail platforms.
- **Open-Source Compatibility –** Built using Python, OpenCV, PyTorch, and CP-VTON.

CHAPTER 5: REQUIREMENT ANALYSIS

5.1 Hardware Requirements

The proposed system is designed to run on **mid-range consumer hardware**, with optional enhancements using GPUs for faster deep learning computations.

- **Input Devices**:
  - Digital camera or smartphone camera for capturing user images.
  - Pre-captured product images of garments.
- **Processing Unit**:
  - Minimum: Intel Core i3 / AMD Ryzen 3.
  - Recommended: Intel Core i5 / Ryzen 5 and above.
- **Memory (RAM)**:
  - Minimum: **8 GB** for basic execution.
  - Recommended: **16 GB or higher** for smooth training/inference of models.
- **Graphics Processing Unit (GPU)**:
  - Recommended: NVIDIA GTX 1650 or higher with CUDA support for PyTorch acceleration.
- **Storage**:
  - At least **20 GB** free disk space for model weights, datasets, and output images.
- **Operating System**:
  - Windows 10/11, Ubuntu Linux, or macOS.

5.2 Software Requirements

The system uses open-source libraries and frameworks for deep learning and image processing.

- **Programming Language**: Python 3.8 or above.
- **Frameworks and Libraries**:
  - **PyTorch** – for implementing CP-VTON and deep learning modules.
  - **OpenCV** – for image preprocessing and segmentation.
  - **NumPy** – for matrix operations and numerical computations.
  - **TorchVision** – for model training and dataset handling.
  - **Matplotlib / Seaborn** – for visualization of outputs.
- **Development Environment**:
  - Jupyter Notebook, PyCharm, or Visual Studio Code.
- **Version Control & Collaboration**:
  - Git/GitHub for source code management.

- **Optional Tools**:
  - CUDA Toolkit (for GPU acceleration).
  - Anaconda environment for package management.

5.5 Feasibility Study

A feasibility study assesses the **technical, economic, and operational viability** of the system.

5.5.1 Technical Feasibility

- CP-VTON and GAN-based frameworks have proven effectiveness in academic research.
- Runs efficiently on consumer hardware with GPU support.
- Implemented entirely with open-source tools.

Conclusion: Technically feasible.

5.5.2 Economic Feasibility

- No specialized hardware (like 3D scanners) required.
- Uses free libraries (PyTorch, OpenCV).
- Only basic computing devices and cameras needed.

Conclusion: Economically viable with low cost of deployment.

5.5.3 Operational Feasibility

- Simple workflow: upload image → select clothing → view try-on result.
- Can be integrated into existing e-commerce platforms with minimal training for end users.

Conclusion: Operationally practical.

CHAPTER 6: MODULE DESCRIPTION

6.1 Modules

6.1.1 Input Module

- **Role:** Collects user and garment images as the primary inputs.
- **Input:**
  - User photograph (front-facing).
  - Clothing image (catalogue image with transparent/white background).
- **Process:**
  - Validate file formats (JPEG, PNG).
  - Standardize image resolution for preprocessing.
- **Output:** Clean input images ready for preprocessing.

6.1.2 Preprocessing Module

- **Role:** Prepares images for downstream tasks.
- **Input:** Raw user image + clothing image.
- **Process:**
  - Resize images (e.g., 256×192 pixels).
  - Normalize pixel values.
  - Remove background noise if needed.
- **Output:** Normalized user and garment images.

6.1.3 Pose Estimation Module

- **Role:** Extracts body key points to define posture and orientation.
- **Input:** Pre-processed user image.
- **Process:**
  - Apply pose estimation (OpenPose/MediaPipe).
  - Detect skeletal key points (shoulders, hips, knees, elbows, etc.).
- **Output:** Skeleton/pose map of user body.

6.1.4 Human Segmentation Module

- **Role:** Divides the user's body into different regions for precise garment overlay.
- **Input:** User image + skeleton map.
- **Process:**
    - Apply human parsing algorithms to separate torso, arms, and background.
    - Generate masks for regions of interest (ROI).
- **Output:** Segmented user image with region-specific masks.

6.1.5 Garment Processing Module

- **Role:** Prepares clothing images for alignment and warping.
- **Input:** Clothing image.
- **Process:**
    - Isolate garment from background.
    - Resize and scale garment proportionally to user's body dimensions.
- **Output:** Clean garment image prepared for warping.

6.1.6 Garment Warping Module (CP-VTON Core)

- **Role:** Warps garment to fit user's body shape and pose.
- **Input:** Garment image + skeleton map + segmentation mask.
- **Process:**
    - Use Geometric Matching Module (GMM) from CP-VTON to align garment with body posture.
    - Preserve garment texture, fabric, and design details.
- **Output:** Warped garment image aligned to user's body.

6.1.7 Image Synthesis Module

- **Role:** Generates final try-on output by blending garment and user image.
- **Input:** Warped garment + segmented user image.
- **Process:**
    - Apply deep generative networks to merge clothing naturally with user photo.
    - Adjust lighting, shading, and boundaries for realism.

- **Output:** Photo-realistic try-on image.

6.1.8 Smoothing & Refinement Module

- **Role**: Enhances realism and corrects artifacts.
- **Input:** Synthesized image.
- **Process:**
    - Apply post-processing filters.
    - Refine garment boundaries and remove blending errors.
- **Output:** Polished, high-quality try-on image.

6.1.9 Output & Visualization Module

- **Role**: Displays final results to user.
- **Input**: Final synthesized try-on image.
- **Process**:
    - Present output in a user interface.
    - Allow saving/exporting of results.
- **Output:** Final try-on image visible to the end user.

6.2 Module Integration

The operation of VCTOS can be represented as:

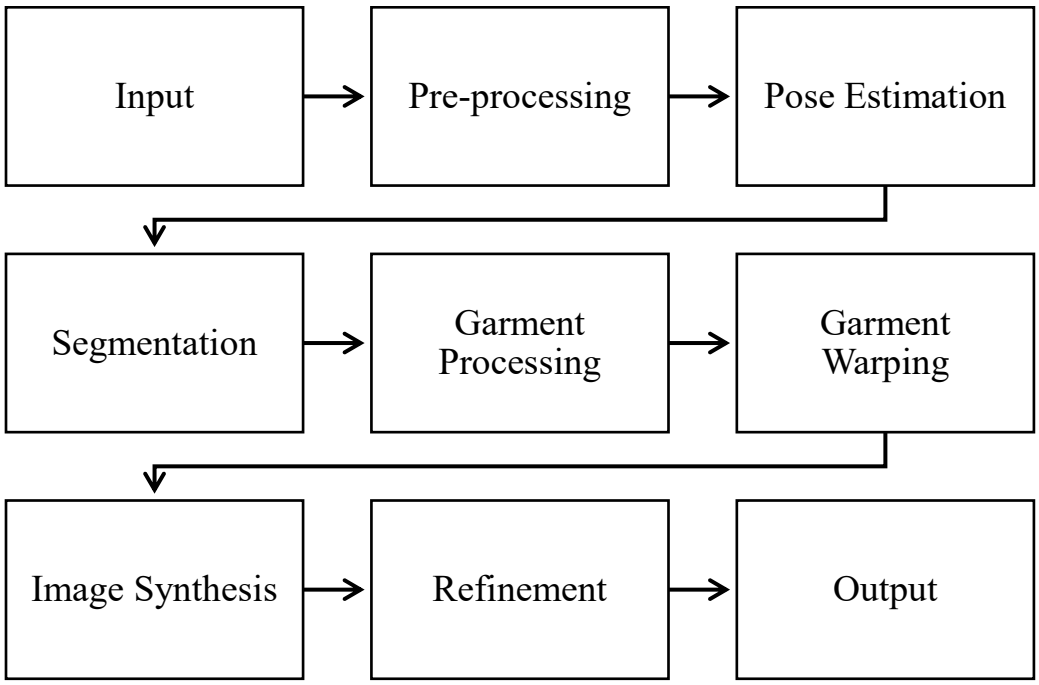


Fig.1: Module Flow Diagram

This pipeline ensures smooth data flow and guarantees **photo-realistic try-on experiences**.

CHAPTER 7: DATA FLOW DIAGRAM

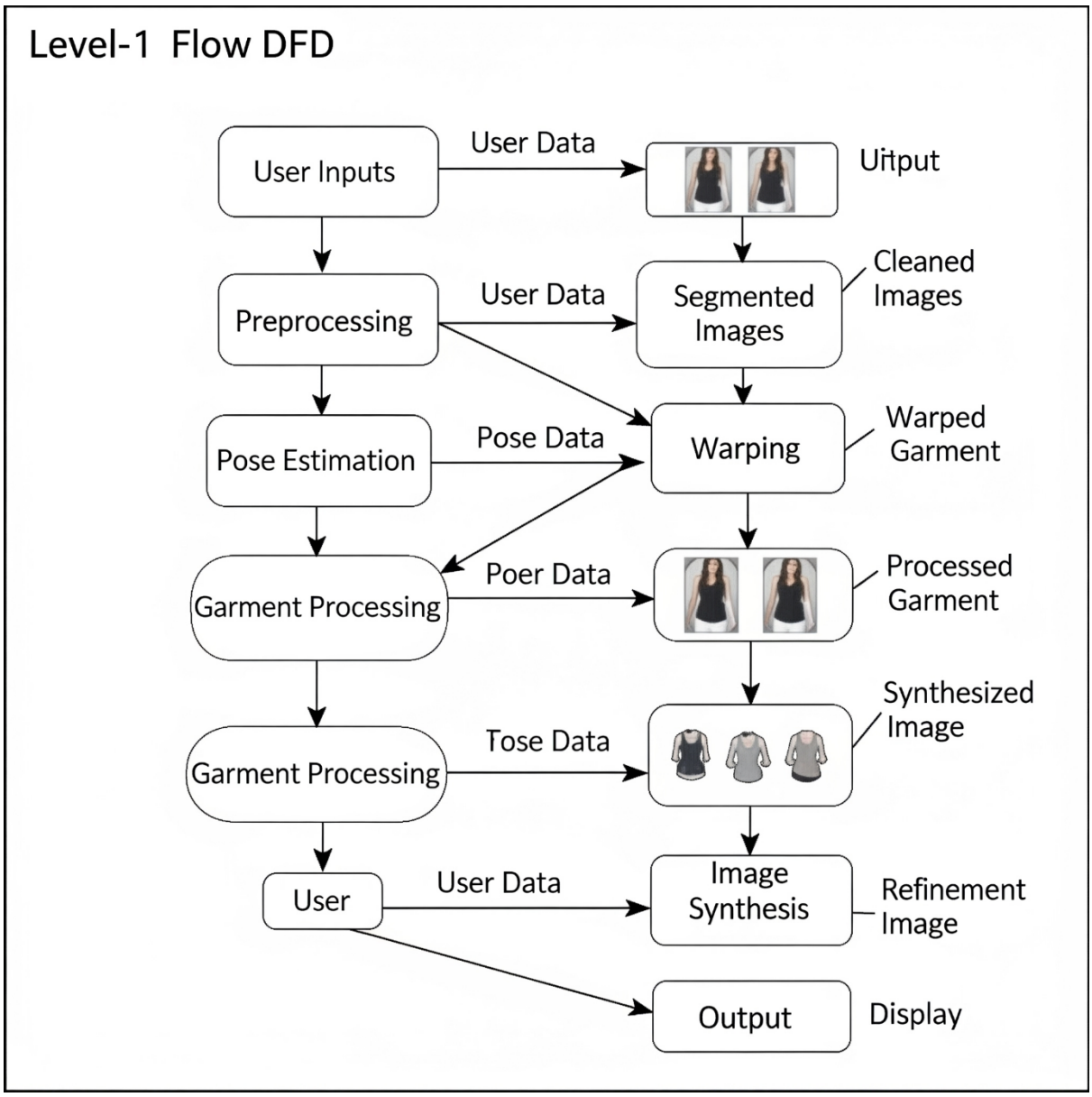


Fig.2: Level-1 Data Flow Diagram

CHAPTER 8: APPLICATIONS, ADVANTAGES, LIMITATIONS & FUTURE SCOPE

8.1 Applications

8.1.1 E-Commerce Platforms

- Provides customers with a **realistic preview** of garments before purchase.
- Reduces return rates caused by incorrect size or dissatisfaction with fit.
- Improves consumer confidence, leading to increased conversion rates.

8.1.2 Fashion Design & Retail

- Enables designers to **digitally showcase collections** without manufacturing physical samples.
- Supports fashion houses in conducting **virtual fashion shows**.
- Retailers can update catalogues with virtual models wearing new garments quickly.

8.1.3 Consumer Personalization

- Allows users to **try different styles and outfits** virtually.
- Helps customers choose clothing combinations that suit their body type and preferences.

8.1.4 Marketing & Advertising

- Retailers can create engaging campaigns where users can upload their photos and try products.
- Social media platforms can integrate try-on filters for promotional campaigns.

8.1.5 Education & Research

- Provides a platform for research in **computer vision, human parsing, and image synthesis**.
- Can be used in **fashion technology courses** to teach virtual garment design.

8.1.6 Sustainability in Fashion

- Reduces the need for producing physical samples.
- Minimizes returns and logistics waste, contributing to **eco-friendly fashion retail**.

8.2 Advantages

1. **Customer Confidence**
   - Realistic previews give customers assurance about style and fit.
2. **Cost-Effective**
   - Uses open-source frameworks (PyTorch, OpenCV, CP-VTON).
   - Eliminates the need for expensive 3D scanners or specialized hardware.
3. **Scalability**
   - Easily integrable into existing e-commerce platforms.
   - Can support multiple garment categories (tops, dresses, jackets).
4. **Accessibility**
   - Works with basic consumer images captured by smartphone or webcam.
5. **Retailer Benefits**
   - Reduces product return rates.
   - Enhances user engagement and loyalty.
6. **Sustainability**
   - Reduces waste from returns and overproduction of physical samples.

8.3 Limitations

1. **Input Image Quality**
   - Results depend heavily on the resolution and clarity of user and garment images.
2. **Pose Constraints**
   - Works best with **frontal, upright photos**; extreme poses may reduce accuracy.
3. **Garment Complexity**
   - Intricate designs, layered clothing, and transparent fabrics are harder to simulate realistically.
4. **Computational Requirements**
   - CP-VTON and GAN-based models require **GPU acceleration** for fast processing.
5. **Realism Issues**
   - Minor artifacts may appear around garment boundaries.
   - Lighting inconsistencies between user and garment images can reduce realism.

6. **Limited Interaction**
    - Current prototype supports image-based try-on; **real-time AR/VR support** requires further research.

8.4 Future Scope

8.4.1 Real-Time Try-On

- Extend system for **live try-on experiences** using webcams or mobile cameras.
- Integrate with augmented reality (AR) for interactive shopping.

8.4.2 Advanced Garment Handling

- Improve handling of **layered, folded, and transparent garments**.
- Enhance texture preservation using advanced **GAN and diffusion models**.

8.4.3 Personalization

- Introduce user-specific profiles that account for **body measurements**.
- Enable AI-based **style recommendations** based on past preferences.

8.4.4 Cross-Platform Integration

- Extend to **mobile apps, VR headsets, and smart mirrors**.
- Integrate with leading e-commerce platforms like Amazon, Myntra, and Flipkart.

8.4.5 Research & Development

- Expand dataset diversity to include varied poses, lighting, and garment types.
- Apply the system in **virtual fashion shows and metaverse applications**.

REFERENCES

1. Han, X., Wu, Z., Wu, Z., Yu, R., & Davis, L. S. (2018). **VITON: An Image-based Virtual Try-on Network**. *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, pp. 7543–7552.
2. Wang, B., Zheng, H., Liang, X., Chen, Y., Lin, L., & Yang, M. (2018). **Toward Characteristic-Preserving Image-based Virtual Try-On Network (CP-VTON)**. *Proceedings of the European Conference on Computer Vision (ECCV)*, pp. 589–604.
3. Jetchev, N., Bergmann, U., & Vollgraf, R. (2017). **Texture Synthesis with Spatial Generative Adversarial Networks (SGAN)**. *arXiv preprint arXiv:1611.08207*.
4. Cao, Z., Hidalgo, G., Simon, T., Wei, S. E., & Sheikh, Y. (2019). **OpenPose: Realtime Multi-Person 2D Pose Estimation Using Part Affinity Fields**. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 43(1), 172–186.
5. Ma, L., Jia, X., Sun, Q., Schiele, B., Tuytelaars, T., & Van Gool, L. (2018). **Pose Guided Person Image Generation**. *Advances in Neural Information Processing Systems (NeurIPS)*, 31, 405–415.
6. Goodfellow, I., Bengio, Y., & Courville, A. (2016). **Deep Learning**. MIT Press.
7. OpenCV Documentation. (2024). *Open Source Computer Vision Library*. Retrieved from: https://docs.opencv.org
8. PyTorch Documentation. (2024). *An Open Source Machine Learning Framework*. Retrieved from: https://pytorch.org/docs
9. Amazon Inc. (2022). **Zeekit Virtual Try-On Technology**. Retrieved from: https://www.aboutamazon.com
10. Myntra. (2023). **Myntra Virtual Try-On Initiative**. Retrieved from: https://www.myntra.com
11. Microsoft Azure AI. (2023). **AI-Powered Fashion & Retail Solutions**. Retrieved from: https://azure.microsoft.com/solutions/ai
12. Dahiya, S. (2021). **Computer Vision in Fashion Retail: A Survey of Applications and Techniques**. *Journal of Retail Technology Research*, 15(3), 201–215.